

Experienced automotive engineer specializing in chassis controls and by-wire systems with 5+ years in MiL, HiL and in-vehicle validation. Skilled in systems integration and dedicated to driving innovation through technology.

Education

- **Master of Science in Automotive Engineering** - (Major: Vehicle Dynamics & Controls) Aug '23 - Aug '25
Clemson University - International Centre for Automotive Research Greenville, SC, US
Relevant Coursework: Vehicle dynamics, Automotive stability & safety systems, Classical & Modern control theories, Automotive control systems design, Robust predictive control and Applied systems & electronics integration
- **Bachelor of Engineering in Mechanical Engineering** Aug '14 - May '18
SSN College of Engineering Chennai, Tamil Nadu, India

Professional Experience

Chassis Controls Software & Systems Engineer May '25 - July '25
Ford Motor Company - Research & Advanced Engineering Division Dearborn, MI, US

- Led dual-redundant Steer-by-Wire system development; executed real-time validation
- Developed and integrated control software; validated via MiL, HiL, in-vehicle testing; maintained GitHub version control
- Engineered hardware setup; established CAN communication; performed system testing/calibration at Ford proving grounds; ensured reliability and robustness

Deep Orange 16 Intern - Vehicle dynamics & controls engineer Jan '24 - Apr '25
Clemson University (ICAR), Greenville, Greenville, SC, US

- Designed and implemented HiL test setup for low-voltage **Brembo Sensify brake-by-wire** and **Fox Live Valve suspension** systems; developed brake supervisory control and start-up state machine on New Eagle ECU with MiL/HiL validation; created and executed hardware test plans, calibration, fault detection and fail-safe strategies.
- Developed **Direct Yaw Moment Control (DYC)** algorithm for torque vectoring in High-Voltage 4-wheel independent hub motor series hybrid vehicle to enhance stability and emulate differential behavior.
- Built custom **HMI on MoTeC C1812** display console; integrated CAN communication and GUI for real-time vehicle data visualization; enabled driver interaction via steering wheel navigation buttons

Graduate Research Assistant Mar '24 - Dec '24
Clemson University (ICAR) - Dr. Matthias J. Schmid Greenville, SC, US

- Served as data acquisition and VD controls test engineer for Deep Orange 14 U.S. Army project; instrumented BMW 535xi for dynamic and brake testing (4-post shaker, skid pad, slalom, double lane change) at in-house and ITIC proving grounds

Assistant Manager, Product Development - Chassis testing & validation Aug '18 - Jun '23
Royal Enfield (A unit of Eicher Motors Pvt. Ltd.), Chennai, Tamil Nadu, India

- Managed chassis systems integration, resolving development issues through cross-functional collaboration, root cause analysis, and validated design changes; established testing procedures meeting ECE, FMVSS, and Indian standards.
- Led ABS development and vehicle performance testing, driving calibration, validation, and DVP execution for braking, ride, and handling; hands-on experience in new prototype development process (M1, VB & TT) and assembly.

Graduate Projects

- Developed predictive dual-loop HVAC thermal management system for EV battery and cabin using MPC; optimized compressor, pump, and blower to balance cooling, enhance energy efficiency, extend battery life, and improve passenger comfort.
- Designed hierarchical model-based controller using Direct Yaw Moment Control (DYC) for optimal torque vectoring in four-wheel independent motor vehicle; validated stability improvement through simulations of double lane change, J-turn, and sine-with-dwell maneuvers on varied surfaces.
- Developed and implemented Adaptive Cruise Control and Lane Keeping on a RC car using Arduino; integrated ultrasonic sensors with Kalman filter sensor fusion, PID tuning, and conducted data-driven testing on random tracks.

Skills

- MATLAB Embedded Coder, Simulink, Stateflow, CarSim, dSPACE, ATI Vision, VBox Automotive, Speedgoat, HBM eDAQ, CANalyzer, GitHub (Version Control), Embedded C/C++, Microsoft 365 Suite